9.4 Practical circuits

In this topic you will:

- draw circuit diagrams that include symbols for cells, switches, resistors, variable resistors, ammeters, voltmeters, lamps and buzzers
- make circuits that include some or all of these components.

Getting started

Work in pairs.

1 Draw the circuit symbol for each of these: cell, switch, resistor, ammeter, voltmeter, lamp and buzzer.

2 Beside each circuit symbol, write the function of each component.

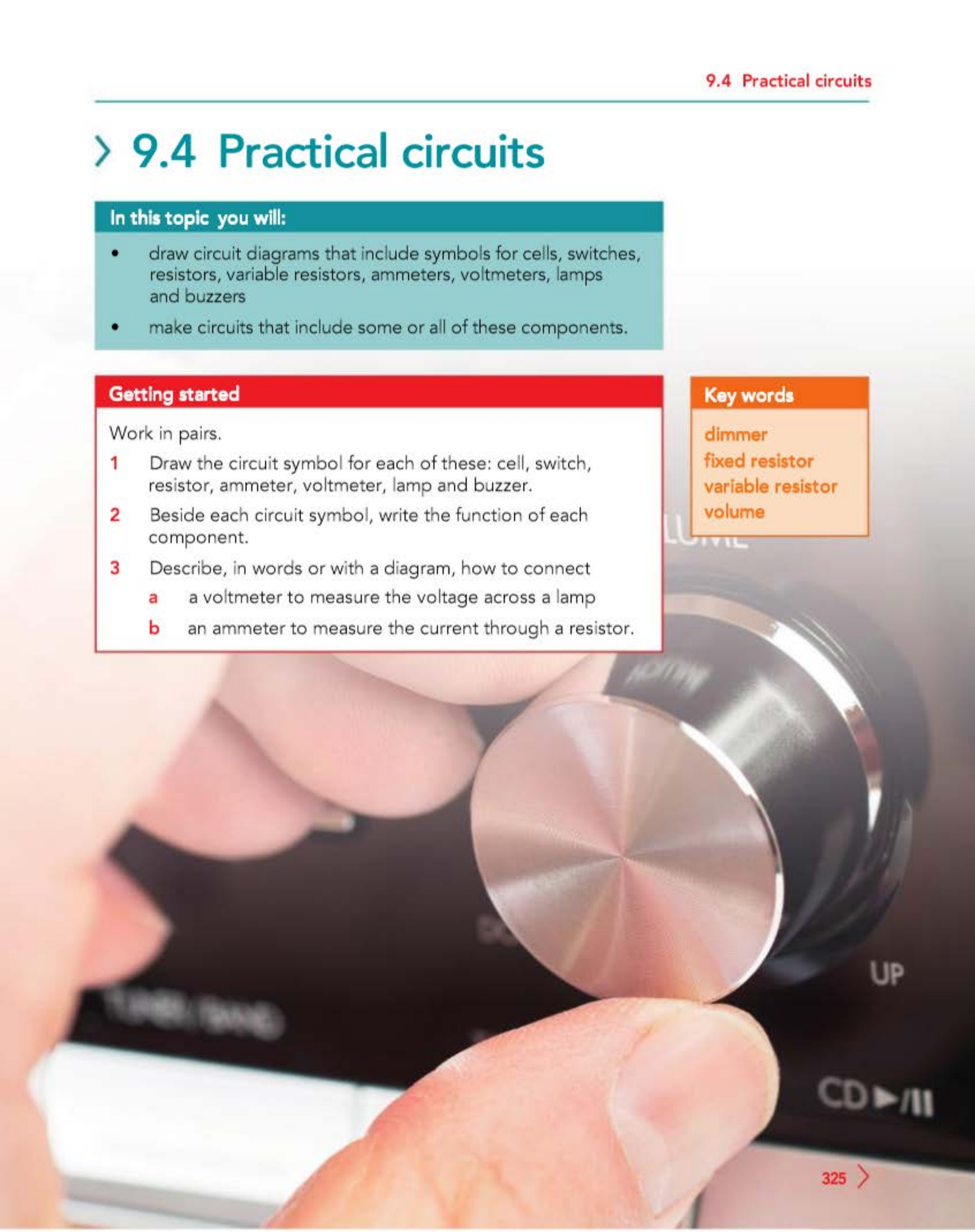
3 Describe, in words or with a diagram, how to connect

a a voltmeter to measure the voltage across a lamp

b an ammeter to measure the current through a resistor.

Key words

dimmer
fixed resistor
variable resistor
volume



Variable resistors

In many circuits, it is useful to be able to change the resistance. For example, in a **dimmer** switch for a lamp, or in the **volume** control of a music player. In this case, the word volume means loudness of sound.

The component that is used to change resistance is called a **variable resistor**.

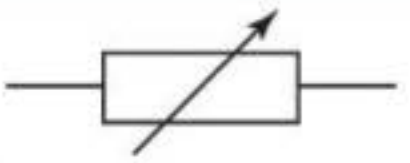


These light switches (called dimmer switches) contain variable resistors that can be turned to control the brightness of lamps in a room.



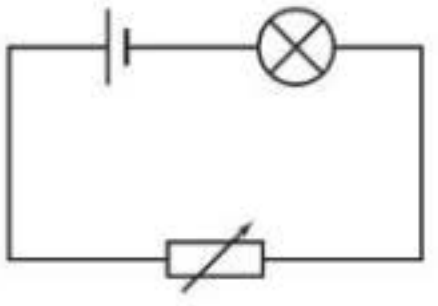
Sliding controls at the bottom of this sound recording equipment are variable resistors for adjusting volume.

The circuit symbol for a variable resistor is similar to that for a **fixed resistor**, but with an arrow through the symbol.



The term fixed resistor is sometimes used for the type of resistors you learnt about in Topic 9.3, to avoid confusion with variable resistors.

This diagram shows how a variable resistor is used to control the brightness of a lamp.



As the resistance of the variable resistor is increased, the current in the circuit decreases. A smaller current flowing through the lamp makes the lamp dimmer.

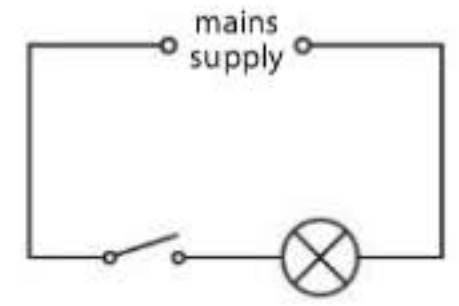
As the resistance of the variable resistor is decreased, the current in the circuit increases. A larger current flowing through the lamp makes the lamp brighter.

Everyday circuits

Electric circuits are used in many homes, schools, cars, bicycles, phones, music systems, watches, computers and lots more things that we use every day.

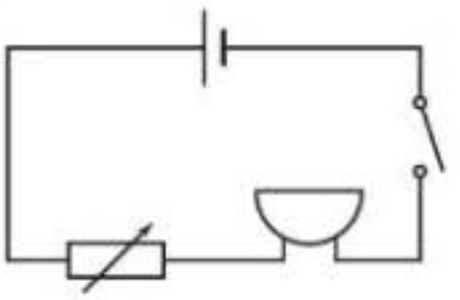
Scientists are always designing new circuits, many of which make life easier for us. Here we will look at some examples.

When opening the door of a refrigerator (fridge), a light comes on inside. The circuit for this is shown in the diagram.

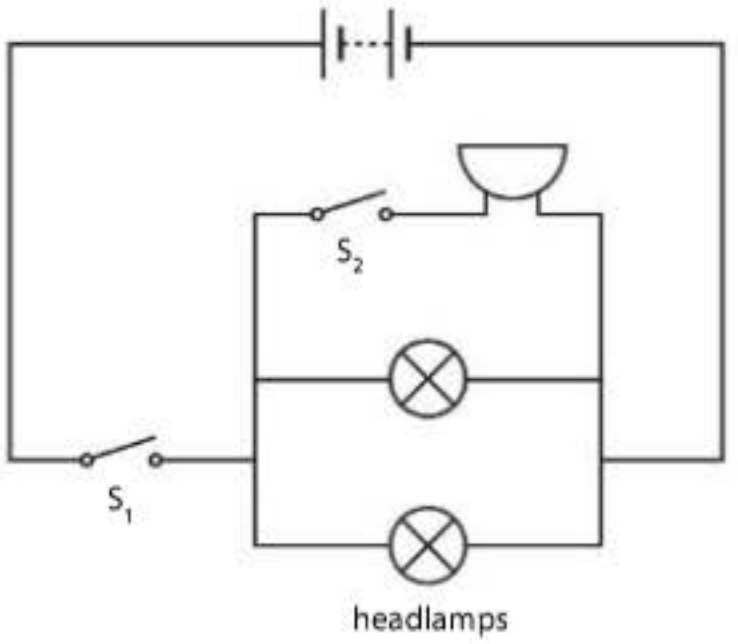


The switch is opened and closed by the movement of the fridge door. When the door is open, the switch is closed and when the door is closed, the switch is open.

Some equipment is fitted with buzzer to make a warning sound if something is done incorrectly. Often, the loudness of the sound can be adjusted. This circuit shows how this is done.



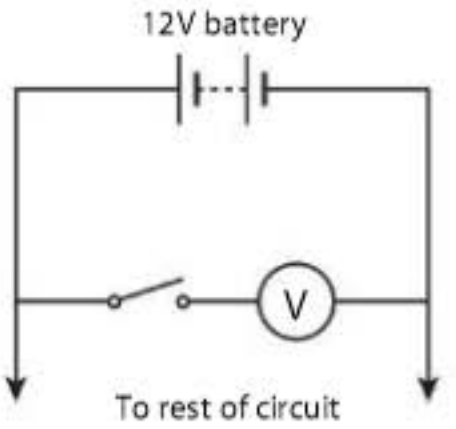
Some cars have a buzzer that makes a sound when the driver opens the door while the headlamps are switched on. The door of the car controls the switch for the buzzer, but the buzzer should only operate when the headlamps are on.



The headlamps are connected in parallel, so they can be switched on and off together. The buzzer is part of the same branch of the parallel circuit, as shown in the diagram.

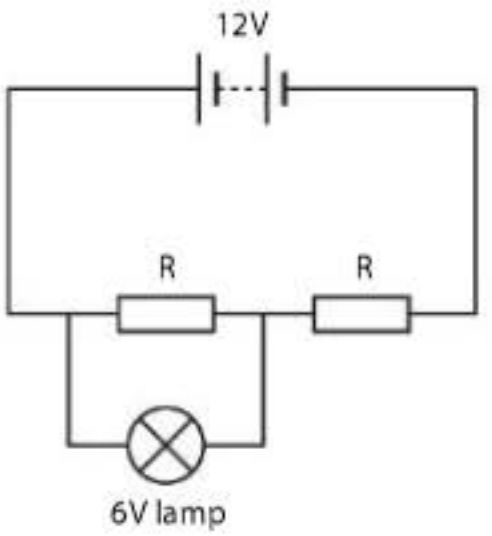
Switch S_1 controls the headlamps. Switch S_2 is controlled by the door. When the door is open, the switch is closed and when the door is closed, the switch is open. Current cannot flow to switch S_2 unless switch S_1 for the headlamps is closed. That means a buzzer will sound when you open the door while the headlamps are on.

Equipment that uses 12 V batteries include cars, trucks and some boats. These batteries need to be recharged from time to time. Sometimes, a battery voltage indicator is fitted so the voltage of the battery can be measured.



The circuit diagram shows how a switch can be used to display the battery voltage. A voltmeter is connected in parallel across the battery, and a switch is included in this branch of the parallel circuit.

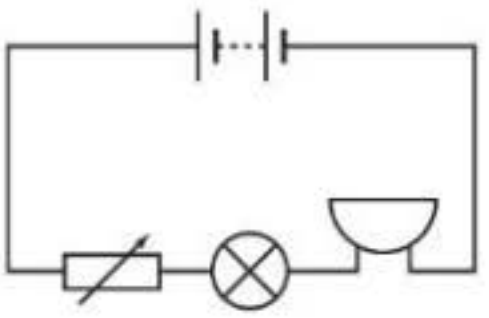
Sometimes a lamp with a low voltage rating is required in a circuit with a battery of higher voltage. For example, a 6 V lamp can be operated using a 12 V battery. Resistors are used to produce different voltages. If two resistors of equal resistance are connected in series with a battery, then the battery voltage will be shared equally across each resistor. A 6 V lamp can then be safely connected in parallel with one of the resistors, as shown. The lamp will receive a 6 V supply and work properly.



Questions

1 Draw the circuit symbol for
   a a fixed resistor
   b a variable resistor.

2 Look at the circuit diagram.



What can this circuit be used for? Write **one** letter.

   A Changing the loudness of the buzzer, keeping the brightness of the lamp constant.
   B Changing the brightness of the lamp, keeping the sound from the buzzer constant.
   C Changing the loudness of the buzzer and brightness of the lamp at the same time.
   D Changing the loudness of the buzzer and brightness of the lamp separately.

3 A circuit will use one cell and two lamps.

One switch must turn both lamps on and off at the same time.

The brightness of each lamp must be controlled separately.

Draw a circuit diagram for this circuit.

4 A circuit has two cells, a lamp and a buzzer in series.

a Draw a circuit diagram to show how the current through the buzzer and the voltage across the buzzer could be measured at the same time.

b Draw **one** circuit diagram to show how:

- the lamp and the buzzer can be switched on and off separately
- the brightness of the lamp can be changed without affecting the sound from the buzzer.

Think like a scientist

Designing and building circuits

In this activity, you will draw circuit diagrams and then build the circuits.

Work in pairs.

You will need:

- cells, lamps, buzzers, variable resistor, voltmeter, ammeter, selection of fixed resistors, wires, connectors.

Method

1 Write a risk assessment for building electric circuits.

2 Draw circuit diagrams for the following circuits.

Circuit 1 • two lamps can be switched on and off separately • the current through one of the lamps can be measured	**Circuit 2** • two lamps can be switched on and off together using one switch • the brightness of one of the lamps can be varied
Circuit 3 • a fixed resistor and a variable resistor in series • the voltage across the variable resistor can be measured	**Circuit 4** • one lamp can be switched on and off • a buzzer can only be switched on at the same time the lamp is on • the brightness of the lamp can be varied without affecting the sound from the buzzer

3 Ask your teacher to check your risk assessment and circuit diagrams, and then build each of the circuits.

Continued

Questions

1 Did all your circuits work as you expected? Describe any observations that you did not expect.

2 Describe any improvements you could make to your circuits. You can suggest the use of different equipment. You should describe this equipment, rather than suggesting 'better' equipment.

Summary checklist

- ☐ I can draw the circuit symbol for a variable resistor.
- ☐ I know how variable resistors are used in circuits.
- ☐ I can compare circuit diagrams that contain cells, switches, fixed resistors, variable resistors, ammeters, voltmeters, lamps and buzzers.
- ☐ I can build circuits that contain cells, switches, fixed resistors, variable resistors, ammeters, voltmeters, lamps and buzzers.

